

Light Verbs, Heavy Meaning?

Exploring Informativity, Frequency and Morpho-Syntactic Interface of Simplex Verbs and ‘root+hata’ Light Verb Constructions in Korean

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1 Introduction

1.1 Motivation and Background of the Study

Polinsky and Magyar (2020) propose that head-final languages such as Korean favor low-informativity simplex verbs, expanding the verbal lexicon instead via light verb constructions. Their study measures the verb-to-noun ratio across a 33-language sample; this paper instead tests further predictions that follow from their informativity account but that they do not themselves test — namely, that low-informativity simplex verbs should also be more polysemous, less self-similar across usage contexts, and more frequent than the nominal component of comparable LVCs. Polinsky & Magyar’s own use of ‘polysemous’ refers to categorial polyvalence — root-level noun/verb ambiguity without derivation, as found in Tongan, Mandarin Chinese, Riau Indonesian, and Mapudungun — not semantic polysemy in the ordinary sense, and Korean is not among the languages they characterize this way. The present paper instead investigates semantic polysemy: whether a word’s meaning is discretely multiple (dictionary sense count) and/or contextually broad rather than narrow (embedding self-similarity, following Ethayarajh (2019)).

This paper will be structured as follows: Section 2 provides a theoretical framing of Idsardi and Raimy (2013)’s three-way distinction between types of linearization and an overview of linearization phenomena in Korean sample verbs, followed by Section 3, which details the methodology, including our within-pair comparative design and specific metrics for evaluating dictionary polysemy, embedding self-similarity, and corpus-derived frequency. Section 4 presents the empirical results across the five pairs, and Section 5 discusses these findings in light of the proposed informativity and morpho-syntactic hypotheses. Finally, Section 6 offers concluding remarks, limitations, and directions for future research.

1.2 Research Questions and Hypotheses

- RQ: Are Korean’s simplex verbs more polysemous, less contextually stable, and more frequent than the comparable ‘-hata’ constructions, as predicted by their lower informativity

in a head-final grammar?

- Hypothesis: Simplex verbs in Korean are predicted to be more polysemous, less self-similar across usage contexts in embedding space and more frequent in Corpus usage than the Light Verb Constructions.

2 Theoretical Background and Related Works

2.1 Linearization in Korean

This paper examines whether Idsardi and Raimy (2013)’s three-way distinction between types of linearization—Immobilization, Vocabulary Insertion, and Serialization—can be shown to apply generally to Korean word-formation, using the root+*hata* pattern and related morphology as a testing ground.

(i) Immobilization. In Korean, this operation uniformly governs both complex predicates and affixation: an abstract verbalizer merges hierarchically with a nominal root and is immobilized into the strict [root] + [*hata*] string, while a causative or passive head merging with a verbal root is likewise fixed before the final ending *-ta* (e.g., 먹다 *mekta* eat’ → 먹이다 *mekita* feed’; 잡다 *capta* catch’ → 잡히다 *caphita* be caught’) to form a complex V^0 before Vocabulary Insertion.

(1) a.	먹다 mek-ta eat-DECL ‘eat’	→	먹이다 mek-i-ta eat-CAUS-DECL ‘feed’
b.	잡다 cap-ta catch-DECL ‘catch’	→	잡히다 cap-hi-ta catch-PASS-DECL ‘be caught’

This Immobilization is further corroborated by a head-marking asymmetry: inflectional morphology (tense, mood) attaches strictly to *hata* itself (생각했다, 노래했다, 토했다), never to the nominal root (생각, 노래, and 토 never inflect independently), whereas simplex verbs like *heyalita* inflect directly on the lexical root (헤아렸다) in the absence of a distinct light-verb layer.

(2) a.	헤아렸다 heyaly-ess-ta consider-PST-DECL ‘considered’	vs.	생각했다 sayngkak-hay-ss-ta think-do-PST-DECL ‘thought’
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(ii) Vocabulary Insertion. Korean shows two clear cases of this. First, light-verb alternation: the same abstract voice head is spelled out as 하다 (*hata*, active), 되다 (*toyta*, passive/inchoative), or 시키다 (*sikhita*, causative) depending on which voice features it carries in a given root+*hata* base (e.g., 설명하다 *selmyenghata* ‘explain’ / 설명되다 *selmyengtoyta* ‘be explained’ / 설명시키다 *selmyengsikhita* ‘cause to explain’) — three Vocabulary Items competing to realize one and the same terminal node under different conditions.

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|---|--|
| <p>(3) a. 설명하다
selmyeng-ha-ta
explanation-do-DECL

‘explain’ (active)

설명시키다
selmyeng-sikhi-ta
explanation-cause-DECL

‘cause to explain’ (causative)</p> | <p>설명되다
selmyeng-toy-ta
explanation-become-DECL

‘be explained’ (passive/inchoative)</p> |
|---|--|

Second, the causative/passive suffix discussed above shows the same kind of competition in Korean: which form surfaces — *hi/i/li/ki* (e.g., 잡히다 *cap-hita*, 먹이다 *mekita*, 날리다 *nallita* ‘make fly’, 안기다 *ankita* ‘be held’)¹ — depends purely on which root it attaches to, the same root-conditioned competition familiar from English past-tense allomorphy (*-ed/-t/ablaut*, e.g. *walked* vs. *sang*), just operating over one small grammatical piece inside the word rather than the whole word.

- | | |
|--|---|
| <p>(4) a. 먹이다
mek-i-ta
eat-CAUS-DECL

‘feed’

잡히다
cap-hi-ta
catch-PASS-DECL

‘be caught’</p> | <p>날리다
nal-li-ta
fly-CAUS-DECL

‘make fly’

안기다
an-ki-ta
hug-PASS-DECL

‘be held’</p> |
|--|---|

(iii) Serialization. In the case of Korean irregular verb paradigms, the surface string cannot be derived via simple string concatenation of stem and affix, requiring late repairs²

¹*hi/i/li/ki* are the suffixes Korean uses to mark causative or passive voice. Which reading a given form receives is determined by argument structure, not by which suffix appears: a causative reading introduces an additional causee argument that surfaces as an object (e.g., 아이를 안기다 ‘have [someone] hold the child’), while a passive reading has no such object (e.g., 아이가 안기다 ‘the child is held’).

²Korean orthography reflects this asymmetry: *l*-final stems are covered by the general clause of Korean Orthography Article 18 (National Institute of Korean Language, 2017), whereas the *p*, *t*, and *lu* alternations are each listed as separate supplementary clauses 1–3 explicitly labeled irregular conjugation and apply only to a restricted class

(5)	a.	<i>l</i>-deletion:	*살-ㄴ다 *sal-n-ta *live-PRES.DECL (ill-formed)	→	산다 sa-n-ta live-PRES.DECL '(s/he) lives'
	b.	<i>p</i>-alternation:	*돕-아 *top-a *help-INF (ill-formed)	→	도와 tow-a help-INF 'helping'
	c.	<i>t</i>-alternation:	*걷-어 *ket-e *walk-INF (ill-formed)	→	걸어 kel-e walk-INF 'walking'
	d.	<i>lu</i>-alternation:	*모르-아 *molu-a *not.know-INF (ill-formed)	→	몰라 moll-a not.know-INF 'not knowing'

This serialization mechanism connects directly to the head-marking analysis in (i): *hata* itself requires late phonological repair. When combining with vowel-initial affixes, it surfaces irregularly as *haye* (하여) or contracted *hay* (해) rather than concatenative **ha-a* (e.g., 생각했다 *sayngkakhayssta*, not **sayngkakhassta*). Because *hata* is the functional head that carries inflection up to T across all target complex predicates, every inflected root+*hata* form in this study inherently undergoes non-concatenative serialization at this shared morpheme.

2.2 Headedness of Korean

Korean is rigidly head-final across all major syntactic domains: modifying clauses precede the noun, adpositions follow their complements, and verbs occur clause-finally in both root and non-finite clauses (Polinsky and Magyar, 2020):

(6) a. Noun Phrase 철수-가 헤아리-ㄴ 마음
Chelswu-ka heyali-n maum
Chelswu-NOM consider-ADN.PST mind/feelings
‘the feelings that Chelswu considered’

of verbs (e.g., *pat-ta* ‘receive’ stays *pat-a*, unlike *ket-ta* ‘walk’ → *kel-e*). These alternations are widely thought to originate historically from regular phonetic processes, though they no longer apply productively in the synchronic grammar; we remain agnostic on their synchronic phonological or morphological analysis here.

- b. Verb Phrase 마음-을 헤아리-다
maum-ul heyali-ta
mind/feelings-ACC consider-PRS.INF
- ‘to consider (someone’s) feelings’
- c. Postpositional Phrase 부엌-에서
pwuek-eyse
kitchen-LOC
- ‘in the kitchen’
- d. Non-finite Clause 노래-를 부르-면서
nolay-lul pwulu-myense
song-ACC sing-WHILE
- ‘while singing a song’

Unlike category-flexible languages like Mandarin, Korean lacks lexical-category polysemy (Polinsky and Magyar, 2020): nominal roots cannot serve directly as predicates without overt derivation (e.g., *kongpwu* ‘study’ or *yoli* ‘cooking’ require functional *ha-*). In this respect, Korean root+*hata* formation directly mirrors Japanese *suru*, which Polinsky and Magyar (2020) highlight as an engine of verbal lexicon expansion. In both languages, a maximally general light verb combines productively with native, Sino-lexical, and foreign roots without expanding the simplex-verb inventory (e.g., *kwukul-hata* ‘to google’). Rather than an arbitrary cross-linguistic extension, our simplex-versus-*hata* comparison directly evaluates the precise mechanism Polinsky and Magyar describe. Figure 1 illustrates this structural baseline for *nolay-hata* ‘to sing’.

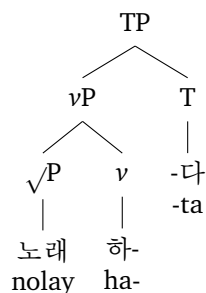


Figure 1: Proposed structure of *nolayhata* ‘sing’, modeled after Adger’s copula analysis: *v* (*hata*) selects a root phrase ($\sqrt{\text{P}}$) directly, with no intervening VP-shell.

Adapting the general architecture of Y.-J. Jung and Miyagawa (2004) — who decompose ditransitive verbal meaning (e.g., *cwuta* ‘give’ as CAUSE+HAVE) across functional heads rather than packing it into a single lexical V — we extend this approach to *hata*: it contributes no thematic content of its own, functioning as a purely grammatical *v*-head. Within Adger (2003)’s taxonomy, *hata* cannot be derived via causative V-to-*v* raising (which requires an independent lexical V) nor via *do*-support (which targets T as a last-resort repair). Instead, we model *hata* after Adger’s analysis of the copula *be*: a non- θ -assigning head realized directly as *v*, selecting a

root phrase ($\sqrt{P}$) as its complement³, with no intervening VP-shell (Figure 1).⁴ We accordingly treat all root+*hata* formations examined here as a unified complex-predicate class, evaluated consistently under this architecture.

3 Methods

3.1 Materials: Minimal Pair Selection

Rather than comparing two independently assembled word lists, this study exploits an internal structural feature of the Korean Swadesh list (Swadesh, 2017). For a small subset of basic concepts, the list records two competing Korean realizations: a native simplex predicate and a root+*hata* (하다) formation. In several cases, one form is explicitly designated as the contemporary “usual” realization, while the other is marked as an older or alternative form.

Candidate pairs were identified by systematically screening the Korean Swadesh list for concepts instantiated by both a simplex predicate and a *hata*-formation sharing equivalent denotational content. This procedure yielded the five concept-controlled minimal pairs presented in Table 1, holding semantic meaning constant while isolating word-formation strategy (simplex versus *hata*-derived).

Concept	Simplex	<i>Hata</i> -derived	Semantic Profile
think (#104)	헤아리다 <i>heyalita</i>	생각하다 <i>sayngkak-hata</i>	Shared denotation (older vs. newer)
sing (#141)	부르다 <i>pwuluta</i>	노래하다 <i>nolay-hata</i>	Shared denotation (older vs. newer)
vomit (#97)	게우다 <i>keywuta</i>	토하다 <i>tho-hata</i>	Shared denotation (older vs. newer)
cook (#235)	굽다 <i>kwupta</i>	요리하다 <i>yoli-hata</i>	Simplex narrower (bake vs. cook)
warm (#180)	덥다 <i>tepta</i>	따뜻하다 <i>ttattus-hata</i>	Adjectival pair (scalar intensity)

Table 1: Minimal pairs of simplex predicates and *hata*-derived formations in Korean.

For the three verbal pairs exhibiting a transparent usual-versus-alternative distribution (think, sing, vomit), the simplex member is hypothesized to pattern consistently across our distributional metrics. The remaining two pairs (cook and warm) are retained for exploratory analysis, though they are not predicted to align symmetrically with the core verbal set: *cook* reverses the expected generality gradient (the *hata*-form is broader rather than narrower), whereas *warm* instantiates

³Although “light verb construction” is standardly described as noun+light-verb in the typological literature, we follow a root-based approach (Marantz, 1997): the same root phrase is not yet categorized as a noun until it merges with a nominalizing head *n*, and combines directly with *v* only in the light-verb construction, bypassing *n* (cf. H. K. Jung 2017, who similarly argues a root phrase ($\sqrt{P}$) is “not yet categorized as a verb” prior to merging with *v*).

⁴*Hata* also exists as an independent main verb ‘do’ (e.g., *swukcey-lul ha-ta* ‘do homework’), which would more plausibly be analyzed as V rather than *v*. However, diagnostics distinguishing this “heavy” use from the “light” verbalizing use in denominal predicates like *sayngkakhata* or *nolayhata* (S. Kim, 1994) yielded unstable judgments when applied to our pairs; we therefore treat *hata* uniformly as *v* across the dataset, which is also required for the Vocabulary Insertion analysis in §?? to go through.

an adjectival scale distinction rather than a stylistic or chronological alternation (see §5).

Structurally, the warm pair (덥다 *tepta* / 따뜻하다 *ttatus-hata*) constitutes an adjectival *root+hata* complex rather than the canonical *Noun+hata* verbal pattern seen in the other four pairs. It is included not as an exemplar of noun-incorporating light verb constructions, but to examine whether the architectural behavior of *hata* as a functional head generalizes beyond verbal LVCs to deadjectival root-licensing.

3.2 Experimental Design and Analytical Measures

Given the controlled nature of the five concept pairs and the intentionally constrained sample size, we adopt a within-pair comparative design rather than aggregating items into unstratified lexical pools. For each pair, the simplex predicate and its *hata*-derived counterpart are evaluated directly across four complementary dimensions: lexicographic polysemy, contextual embedding geometry, speech-based frequency norms, and cross-register corpus distribution.

Dictionary Polysemy. To establish a curated, structural baseline of semantic breadth, sense counts were retrieved programmatically from two authoritative, official lexicographic resources: the prescriptive *Standard Korean Language Dictionary* (*Pyojun Gugeo Daesajeon*) (National Institute of Korean Language, 2024) and the collaborative, open-access dictionary *Urimalsaem* (National Institute of Korean Language, 2016). Query results obtained via public APIs were filtered by part of speech (verbs and adjectives). However, part-of-speech filtering alone does not guarantee lexical relevance, as a single orthographic form can yield multiple etymologically distinct homographs within the same grammatical category—for instance, *kwupta* returns three distinct verbal entries in the standard dictionary, comprising the target thermal/cooking cluster, an intransitive ‘to bend’ sense, and an archaic board-game term. To resolve this ambiguity, we manually identified, for each simplex/*hata* pair, which homograph headword corresponded to the target concept under study (verified against dictionary definitions), and excluded unrelated homographs (e.g., archaic, dialectal, or unrelated senses sharing the same spelling) from the sense count. In practice, the target homograph could be operationalized as the one with the greatest number of documented senses, since peripheral or unrelated homographs consistently carried only one or two entries.

Contextual Embedding Geometry (Self-Similarity). While dictionary sense counts capture discrete polysemy, they do not directly index whether a word’s meaning is broad (spanning many usage contexts with comparatively little semantic drift) or narrow (stable and context-specific)—the generality/specificity distinction. To capture this complementary dimension, we additionally measure semantic distance via contextual embedding self-similarity. Following Ethayarajh (2019), we quantify semantic dispersion through contextual self-similarity, defined as the mean pairwise cosine similarity across contextualized representations extracted from sampled usage instances. Model selection was guided by three criteria: (i) architectural diversity, to ensure that results are not an artifact of a single model’s design; (ii) register diversity in training data, to test whether the observed asymmetries generalize across formal and informal Korean; and (iii) public

availability, to keep the experiment reproducible. To ensure that observed within-pair asymmetries reflect underlying lexical properties rather than idiosyncrasies of tokenization or model architecture, we evaluated three distinct pretrained models spanning diverse training corpora:

- `klue/bert-base`: a standard benchmark trained on curated news, formal texts, and encyclopedic corpora (S. Park et al., 2021).
- `beomi/kcbert-base`: a masked language model trained on unfiltered, user-generated web comments, capturing informal registers and colloquial morpheme combinations (Lee, 2020).
- `text-multilingual-embedding-002`: a multilingual dense retrieval and representation model (Vertex AI), providing an architectural and tokenization counterpoint to monolingual BERT checkpoints.

Corpus Frequency Norms. Spontaneous spoken frequencies were extracted from KoFREN (J.-s. Kim, Choi and Cho, 2024), a large-scale Korean frequency-norm database built upon conversational speech datasets. Because raw token frequencies in natural language are notoriously right-skewed—with high-frequency functional items disproportionately inflating arithmetic means—all frequency metrics were log-transformed ($\log_{10}(\text{freq})$, with zero-count items assigned a log-frequency of 0 by convention) following standard psycholinguistic and distributional practice, preventing extreme values from skewing the paired comparisons.

Cross-Register and Tokenizer Robustness. While KoFREN provides standardized frequency counts based on the Bareun morphological analyzer⁵, the underlying AI Hub⁶ audio transcripts are subject to geographical access constraints, and extensive API-based re-parsing exceeds sustainable open-tier usage limits. To assess whether within-pair frequency patterns persist independently of tokenizer artifacts and speech-specific biases, we conducted a cross-register replication using the Naver Sentiment Movie Corpus (NSMC; L. Park, 2016). Tokens were segmented using the open-source Mecab-ko analyzer (Eunjeon Project, 2013), testing whether the observed lexical distributions generalize to informal, user-generated written discourse.

4 Results

4.1 Dictionary Polysemy

Table 2 reports sense counts for each pair, filtered by part of speech and disambiguated by homograph group (§3.2). STDICT (National Institute of Korean Language, 2024) and *Urimalsaem* (National Institute of Korean Language, 2016) agree exactly on all ten words once homographs are correctly resolved.

Three of the five pairs (sing, cook, warm) show the simplex member as more polysemous by raw sense count, matching the prediction that low-informativity simplex verbs should carry more documented senses than their *hata*-derived counterparts. The remaining two pairs (think, vomit)

⁵<https://bareun.ai/>

⁶<https://aihub.or.kr/>

Concept	Simplex (Senses)	<i>Hata</i> -form (Senses)	Observed Direction
think	<i>heyalita</i> (3)	<i>sayngkakhata</i> (7)	<i>hata</i> > simplex (contrary)
sing	<i>pwuluta</i> (10)	<i>nolayhata</i> (5)	simplex > <i>hata</i> (consistent)
vomit	<i>keywuta</i> (2)	<i>thohata</i> (3)	<i>hata</i> > simplex (contrary)
cook	<i>kwupta</i> (7)	<i>yolihata</i> (2)	simplex > <i>hata</i> (consistent)
warm	<i>tepta</i> (4)	<i>ttattushata</i> (2)	simplex > <i>hata</i> (consistent)

Table 2: Dictionary sense counts by minimal pair. Counts from *Pyojun Gugeo Daesajeon* (STDICT) and *Urimalsaem* align identically following homograph disambiguation. Direction denotes whether the outcome aligns with the hypothesis of simplex breadth (simplex > *hata*).

show the reverse pattern, with the *hata*-form more polysemous. A qualitative examination of the definitions underlying these counts, including two cases where raw sense-count and semantic-breadth judgments diverge, is taken up in §5.

4.2 Contextual Embedding Geometry (Self-Similarity).

Figure 2 reports self-similarity for each word across the five conditions (three embedding models $\times$ two registers) as a heatmap; a dumbbell-plot visualization of the same data, directly comparing simplex and *hata*-form pairs, is provided in the Appendix (Figure 5).

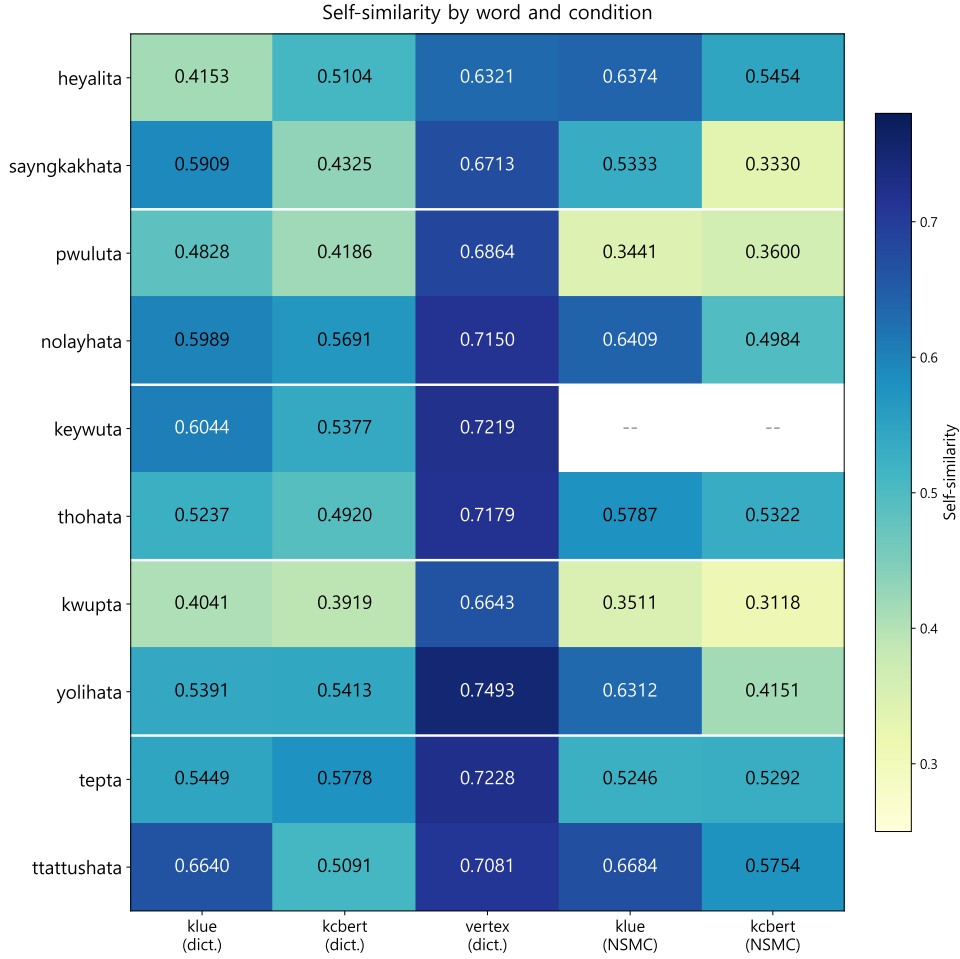


Figure 2: Self-similarity by word and condition (pair boundaries marked by white horizontal lines). *Keywuta* has no NSMC value ($n=1$ example).

Sing and cook are the only two pairs for which the hypothesis holds across every condition tested — all three embedding architectures (klue/bert-base, beomi/kcbert-base, Vertex AI’s multilingual embedding model) and both registers (dictionary examples, NSMC). Vomit shows the opposite pattern: the *hata*-form is less self-similar than the simplex verb in all three conditions for which it could be tested (*keywuta* had too few NSMC occurrences, $n=1$, to compute self-similarity there). Think and warm are model-dependent: klue and Vertex AI agree with the hypothesis for think, while kcbert disagrees in both registers; warm splits along a different line, with klue agreeing in both registers but kcbert and Vertex AI disagreeing in the dictionary-example condition. This same three-way split (sing/cook robust, vomit robustly opposite, think/warm model-dependent) also emerges in the dictionary-polysemy results (Table 2), suggesting it is not an artifact of either method alone.

4.3 Corpus Frequency Norms

Figure 3 plots log-transformed frequency for each pair in both KoFREN (Bareun-tagged spontaneous speech) and NSMC (Mecab-ko-tagged colloquial movie reviews); Table 6 with the underlying values is provided in Appendix.

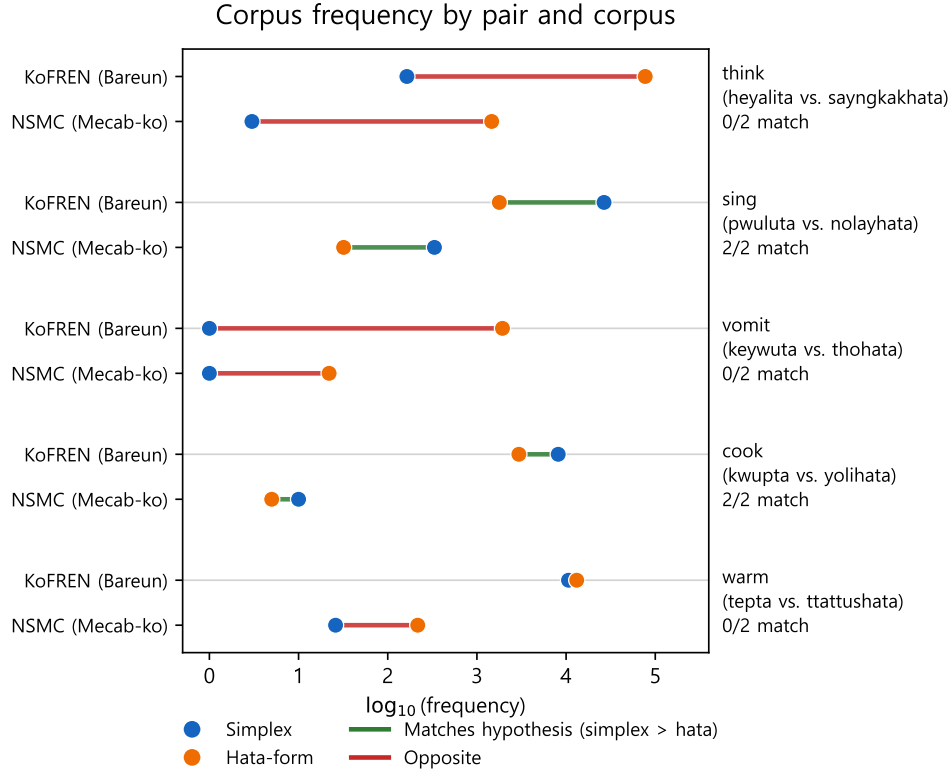


Figure 3: Log-transformed corpus frequency by pair and corpus. Green connects simplex and *hata*-form values where the hypothesis (simplex > *hata*) holds; red where it does not.

Both corpora yield identical directional patterns across all five pairs: *hata*-forms are more frequent in think, vomit, and warm (contrary to the prediction), whereas simplex verbs are more frequent in sing and cook (confirming it). This 5/5 cross-corpus and cross-tokenizer consensus demonstrates that the frequency asymmetry is an intrinsic lexical property rather than an artifact of corpus genre or morphological analysis; yet, only sing and cook align with the informativity account. Notably, *keywuta* is virtually absent in both datasets (zero in KoFREN, one in NSMC), corroborating its lower polysemy, higher self-similarity, and status as an archaic or marginal counterpart to *thohata*.

Summary Across Methods. Table 3 collapses the four methods into a single match/opposite judgment per pair, using the majority direction across the five self-similarity conditions where that method alone is not unanimous.

Concept	Dict. Polysemy	Self-Similarity	KoFREN	NSMC	Total
think	opposite	opposite (2/5)	opposite	opposite	0/4
sing	match	match (5/5)	match	match	4/4
vomit	opposite	opposite (0/3)	opposite	opposite	0/4
cook	match	match (5/5)	match	match	4/4
warm	match	match (3/5)	opposite	opposite	2/4

Table 3: Summary of directional agreement with the hypothesis (simplex more polysemous / less self-similar / more frequent than the *hata*-form) across all four methods.

Two pairs are unanimous in one direction each: sing and cook match the hypothesis on every method and every condition within those methods, while think and vomit oppose it just as consistently. Warm is the one genuinely split case, with the two dictionary-based measures (sense count, and the majority of self-similarity conditions) favoring the hypothesis while both corpus-frequency measures oppose it. This pattern — two pairs behaving as the informativity account predicts, two behaving as a mirror image of it, and one pair divided along a dictionary-vs-corpus line — is examined further in §5, together with the qualitative definitional analysis of semantic breadth and the syntactic status of each *hata*-form.

5 Discussion

5.1 Qualitative Analysis of Lexical Polysemy

A closer examination of the complete sense inventories in the *Standard Korean Language Dictionary* reveals that “semantic breadth” is not a monolithic construct; rather, it bifurcates into at least two distinct linguistic dimensions that do not always align: the raw number of documented senses, and the hypernym/hyponym relationship between the two words’ core definitions — i.e., whether one word’s definition is the more general, superordinate one. Table 4 compares both dimensions across the five pairs: they align in the same direction for only one pair (sing), run in opposite directions for three (think, cook, warm), and are orthogonal for the fifth (vomit, where the two words are near-synonyms with no hypernym relationship despite a difference in raw sense count).

Pair	Raw Sense Count	Hypernym (Definitional Generality)
think	<i>sayngkakhata</i> > <i>heyalita</i> (7 vs. 3)	<i>heyalita</i> > <i>sayngkakhata</i> : one sense of <i>sayngkakhata</i> is defined compositionally via <i>heyalita</i>
sing	<i>pwuluta</i> > <i>nolayhata</i> (10 vs. 5)	<i>pwuluta</i> > <i>nolayhata</i> : its field extends beyond singing to summoning, pricing, and inducing
vomit	<i>thohata</i> > <i>keywuta</i> (3 vs. 2)	<i>thohata</i> = <i>keywuta</i> : near-synonyms differing mainly by era, no hypernym relation
cook	<i>kwupta</i> > <i>yolihata</i> (7 vs. 2)	<i>kwupta</i> < <i>yolihata</i> : <i>yolihata</i> is the culinary superordinate
warm	<i>tepta</i> > <i>ttattushata</i> (4 vs. 2)	<i>tepta</i> < <i>ttattushata</i> : <i>ttattushata</i> has abstract/metaphorical reach <i>tepta</i> lacks

Table 4: Two independent dimensions of semantic breadth: raw dictionary sense count vs. the hypernym/hyponym relationship between the two words’ core definitions. The two agree only for sing; they run in opposite directions for think, cook, and warm; and are orthogonal for vomit, where hypernymy finds no asymmetry despite a raw-count difference.

For the think pair, *heyalita*’s three senses (‘count a quantity’, ‘reach a total’, ‘surmise/infer’) remain anchored to a quantificational or evaluative core, while *sayngkakhata*’s seven senses (‘judge/deliberate’, ‘remember’, ‘intend’, ‘resolve’, ‘imagine’, ‘show care toward’, ‘hold an opinion’) span a considerably richer cognitive and affective space, giving it the larger raw sense count. By hypernymy, however, the relationship reverses: one of *sayngkakhata*’s own senses is defined

compositionally via *heyalita* itself, indicating that the dictionary treats *heyalita* as the more basic, generalizable concept from which at least this sense of *sayngkakhata* is derived.

For the sing pair, *pwuluta*'s ten senses (summon, recite/call roll, dictate, sing, quote a price, chant slogans, induce, provoke, invite, designate) span a broad spectrum of disparate communicative and causative event types, while *nolayhata*'s five senses (sing, birdsong, artistic expression, repetitive nagging, commendation) remain tightly anchored to rhythmic or vocal articulation. Here the two dimensions agree: *pwuluta* is the broader term both by raw sense count and by hypernymy, since its meaning as a general term for vocal or directive utterance subsumes the singing-specific *nolayhata*, rather than the two simply being an older/newer pair for the same core act.

For the vomit pair, *keywuta*'s two senses ('regurgitate food'; 'surrender illicit gains') and *thohata*'s three senses ('expel bodily'; extended to inanimate eruption and to venting emotion) differ in raw count, but the two words are best analyzed as near-synonyms distinguished mainly by register or era, with neither definition framed in terms of the other; we accordingly treat this pair as tied under the hypernymy criterion despite the raw-count asymmetry.

The cook and warm pairs exhibit genuine divergence between the two criteria. In culinary taxonomy, *yolihata* serves as the overarching superordinate term ('prepare food through various cooking processes'), with *kwupta* denoting one specific method of dry-heat cooking; *yolihata*'s only other sense is a colloquial idiom for skillfully handling a person. Yet the cooking-related homograph of *kwupta* alone documents seven senses: roasting/baking food, charcoal-making, ceramic and brick firing, solar salt extraction, heating metal near its melting point, developing a photographic print, and burning a compact disc. By raw count, *kwupta* is thus the more polysemous word, driven by industrial and technical specialization rather than culinary generality; by hypernymy, *yolihata* is the broader, superordinate term.

The warm pair mirrors this dynamic: *tepta*'s four senses (ambient air temperature, physiological body heat, physical object temperature, and a warming property in traditional pharmacology) are strictly literal physical evaluations, giving it the larger raw count. *Ttattushata*'s two senses, however, include an abstract psychological extension (emotional warmth, compassion, and sociability) that *tepta* entirely lacks, making *ttattushata* the hypernym by definitional generality.

These patterns demonstrate that raw dictionary sense counts and hypernym-based definitional generality capture orthogonal axes of lexical breadth, agreeing for only one of five pairs. A native simplex predicate may amass numerous discrete dictionary definitions through concrete, specialized technical uses (as in *kwupta*) while still being the semantically narrower, subordinate term; conversely, a *hata*-formation with fewer sub-senses may nonetheless be the superordinate concept, or the more basic term from which a rival's sense is compositionally derived (as in *heyalita*). Neither lexicographic sense enumeration nor hypernymy — nor, as established in §3.2, contextual embedding dispersion — should be assumed to measure an identical theoretical construct; they constitute complementary rather than interchangeable empirical proxies for lexical breadth.

5.2 Evaluating the Central Hypotheses: Simplex Predicates vs. Light Verb Constructions

We now directly address our core research question: whether Korean simplex verbs are systematically more polysemous, less contextually stable (less self-similar), and more frequent than their root+*hata* counterparts, as predicted by an informativity-driven account of head-final word-formation (Polinsky and Magyar, 2020).

Across the five minimal pairs, the hypothesis is confirmed in 22 of 38 testable comparisons (58%)—holding categorically for two pairs (sing, cook), being categorically reversed for one (vomit), and split across measures for the remaining two (think, warm)—indicating that lower informativity predicts the simplex/*hata* asymmetry as an item-specific tendency rather than a uniform grammatical effect in Korean.

Concept	Simplex / <i>Hata</i> -form	Polysemy	Self-Sim.	Frequency	Overall
think	heyalita / sayngkakhata	× (3 vs. 7)	2/5	× (0/2)	2/8
sing	pwuluta / nolayhata	✓ (10 vs. 5)	5/5	✓ (2/2)	8/8
vomit	keywuta / thohata	× (2 vs. 3)	0/3	× (0/2)	0/6
cook	kwupta / yolihata	✓ (7 vs. 2)	5/5	✓ (2/2)	8/8
warm	tepta / ttattushata	✓ (4 vs. 2)	3/5	× (0/2)	4/8
Total across all pairs and conditions					22/38

Table 5: Hypothesis testing summary: a checkmark/higher ratio means the simplex verb patterns as predicted (more polysemous, less stable, more frequent) than its *hata*-counterpart on that measure.

Table 5 collects all three measures side by side. Note that “polysemy” here denotes semantic sense multiplicity as catalogued in dictionary entries, not the categorial (part-of-speech) polyvalence at stake in Polinsky & Magyar’s original typological criteria (§1), which, as discussed there, does not apply to Korean. The Self-Similarity column reports, out of the model×register conditions available for each pair (up to five: klue/bert-base and kcbert-base on both dictionary examples and NSMC, plus Vertex AI on dictionary examples only), how many show the simplex verb as less self-similar than its *hata*-counterpart; the Frequency column reports, out of the two corpora tested (KoFREN and NSMC), how many show the simplex verb as more frequent.

Taken together, the results do not support a single, consistent answer to the research question: the predicted direction holds in 22 of 38 comparisons overall, but this aggregate figure conceals sharp disagreement across pairs and, for think and warm, across measures within the same pair — two pairs (sing, cook) confirm the hypothesis on every measure tested, one (vomit) reverses it on every measure, and the remaining two are internally split, so the hypothesis itself does not receive a uniform confirmation or disconfirmation, and any conclusion drawn from these data must be conditional on the lexical item rather than general.

Lexicographic Polysemy. By raw sense count (Table 2), the hypothesis holds for sing, cook, and warm (*pwuluta* > *nolayhata*; *kwupta* > *yolihata*; *tepta* > *ttattushata*) and is reversed for think and vomit (*sayngkakhata* > *heyalita*; *thohata* > *keywuta*). However, as established in the qualitative analysis above, raw counts do not track taxonomic or metaphorical generality: for cook

and warm specifically, the *hata*-form’s smaller sense inventory nonetheless includes either the culinary superordinate concept or a genuine abstract extension, while the simplex verb’s larger inventory reflects narrow technical specialization rather than conceptual breadth. Sing is the one pair where the simplex verb is unambiguously broader on both criteria at once.

Contextual Stability and Embedding Self-Similarity. Contextual stability, operationalized as embedding self-similarity, measures representational consistency across usage contexts: lower values indicate greater semantic dispersion. The prediction that simplex verbs are less stable holds robustly and model-independently for sing and cook (5/5 conditions), but reverses consistently for vomit (0/3 conditions; *keywuta* is more stable than *thohata*). Results for think and warm are model-dependent: klue/bert-base and Vertex AI support the hypothesis for think (though kcbert-base and NSMC-evaluated klue disagree), while warm shows mixed outcomes across dictionary and corpus conditions. Overall, cross-condition performance for self-similarity itself is model- and register-dependent for think and warm, in contrast to the fully robust patterns for sing, cook, and vomit.

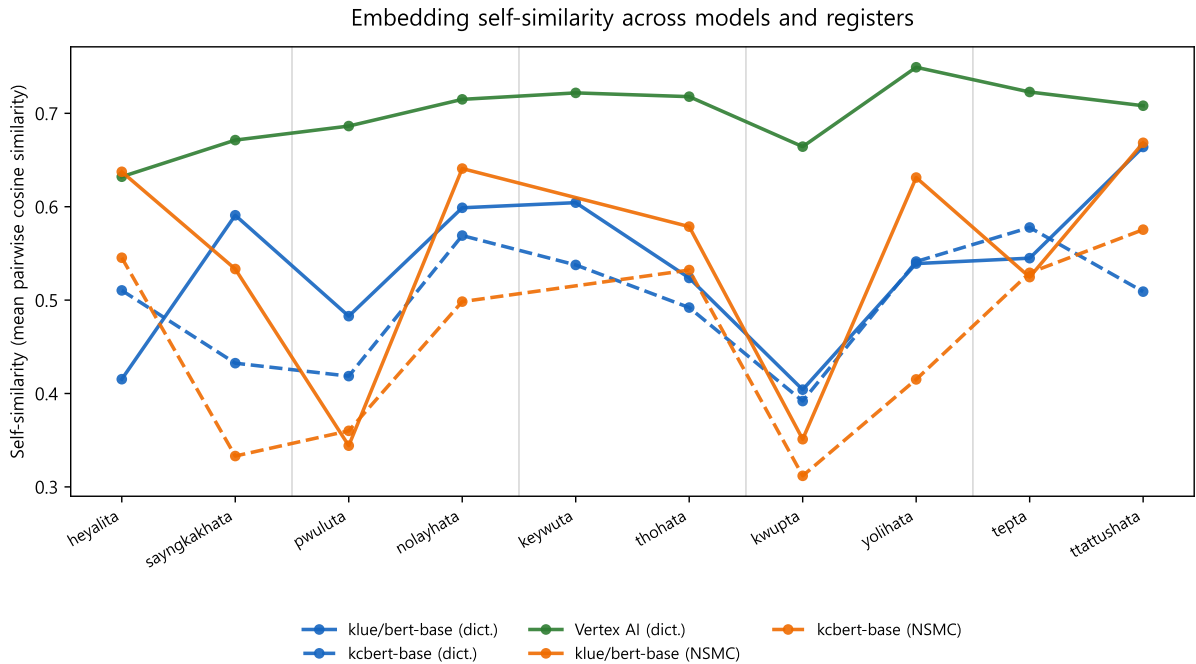


Figure 4: Self-similarity across all five model×register conditions, by word (pair boundaries marked by vertical gray lines).

As shown in Figure 4, Vertex AI exhibits systematically higher absolute self-similarity (0.63–0.75; range 0.12) compared to the BERT-based conditions (0.31–0.67; range 0.19–0.32). However, setting aside two divergent items (*heyalita* and *yolihata*), Vertex AI’s relative word rankings correlate strongly with the other four conditions (pairwise r increases from 0.22 to > 0.90), confirming broad cross-model consensus. Vertex AI’s compressed, elevated values stem from embedding anisotropy (Ethayarajh, 2019)—the crowding of representations into a narrow cone that inflates baseline cosine similarity—likely driven by multilingual training over a shared represent-

ational space. Crucially, this anisotropy affects the entire space uniformly, leaving within-model pairwise comparisons unaffected: both members of each simplex/*hata* pair are evaluated against the same baseline. Thus, while absolute self-similarity cannot be compared directly across architectures, the relative orderings underpinning our hypothesis remain sound.

Corpus Frequency. Across both conversational speech norms (KoFREN) and written discourse (NSMC), the simplex-more-frequent prediction is confirmed for *sing* and *cook* and reversed for *think*, *vomit*, and *warm*, with full agreement between the two corpora on every pair (Table 6). *Sayngkakhata* and *thohata* substantially outstrip the older simplex verbs *heyalita* and *keywuta* in both corpora, and *ttattushata* is likewise modestly more frequent than *tepta* in both.

This agreement across two corpora built from different genres and tokenized by different morphological analyzers makes it unlikely that these particular reversals are simply an artifact of either resource, though a small two-corpus comparison cannot rule out some other property the two datasets happen to share. On this measure, the hypothesis fares worst of the three: a majority of pairs pattern in the direction opposite to what an informativity-driven account predicts.

Theoretical Implications. Polinsky and Magyar (2020) identify two primary strategies by which head-final grammars constrain simplex verb inventories: categorial polyvalence (e.g., Mandarin) and light-verb constructions (e.g., Korean and Japanese). Because Korean lacks categorial polyvalence, our investigation tests the informativity hypothesis specifically through the LVC strategy (with *tepta*–*ttattushata* extending the parallel to adjectival roots).

Across all three empirical measures, the five pairs yield a split outcome: *sing* and *cook* confirm the informativity prediction, *think* and *vomit* reverse it, and *warm* diverges across dictionary and corpus metrics. This mixed distribution argues against a deterministic structural thesis where *hata* categorically yields a broader simplex counterpart. Instead, *hata* acts as a semantically transparent verbalizer; the resulting predicate’s informativity profile is inherited directly from the communicative versatility of its immobilized root (Figure 1). While Polinsky and Magyar’s framework correctly predicts how Korean expands its verbal lexicon without multiplying simplex roots, it does not entail that simplex verbs remain uniformly more polysemous, less stable, or more frequent in the contemporary language.

6 Conclusion

This paper set out to test, on a small but controlled set of Korean minimal pairs, whether the informativity account of head-finality (Polinsky and Magyar, 2020) predicts a systematic asymmetry between native simplex verbs and their semantically related root+*hata* light verb constructions: namely, that the simplex member of each pair should be more polysemous, less contextually stable in embedding space, and more frequent than its *hata*-counterpart. Using five convergent measures — lexicographic sense counts from two independent dictionaries, contextual self-similarity across three embedding models and two registers, and corpus frequency from two independently tokenized corpora — we found that this prediction holds cleanly for two pairs (*sing*, *cook*), is reversed for two others (*think*, *vomit*), and is itself internally split for the fifth

(warm, where dictionary-based measures favor the hypothesis but corpus-based measures oppose it). This item-specific rather than categorical pattern was stable across tokenizers, embedding architectures, and text registers, which suggests it reflects a genuine property of these lexical items rather than measurement noise — but with only five pairs split roughly evenly between confirming and disconfirming the prediction, this is not enough evidence to draw a general conclusion about the hypothesis itself, in either direction.

If a provisional reading is required, the balance of evidence leans slightly toward, rather than against, the informativity prediction (22/38 matching comparisons), but this aggregate is driven almost entirely by two pairs (sing, cook) that confirm it on every measure, while frequency — arguably the measure least confounded by dictionary-editing conventions or model architecture — disconfirms it for a majority of pairs. We do not consider this a reliable basis for either accepting or rejecting the hypothesis.

We therefore stop short of endorsing or rejecting the informativity account on the basis of this study. One plausible reading of the pattern is that *hata* functions as a semantically transparent verbalizing head (Figure 1) whose complex predicate largely inherits the communicative range of the nominal root it combines with, rather than as a device that categorically produces a less informative verbal counterpart to an already-existing simplex verb; but given that the same five measures split three different ways across only five items, this remains a hypothesis for future work rather than a conclusion this study can support. What the data most clearly show is that answering the research question requires a substantially larger and more systematically sampled set of minimal pairs (§??) before any claim about Polinsky & Magyar’s 2020 prediction for Korean — confirming it, reversing it, or reframing it entirely — can be made with confidence.

Limitations

Item Selection and Qualitative Diagnostics. Our analysis relies on only five hand-selected minimal pairs from the Swadesh list, precluding formal statistical hypothesis testing and limiting generalizability across the Korean verbal lexicon. Future work should scale up by systematically extracting productive root+*hata* predicates with simplex counterparts, stratifying them by etymology (native, Sino-Korean, loanword), semantic domain, and argument structure. Furthermore, operationalizing polysemy as raw dictionary sense counts overlooks metaphorical versus literal distinctions, and the syntactic diagnostics (topicalization, pseudo-clefts) rely solely on the first author’s native-speaker intuitions rather than controlled multi-speaker elicitation.

Data Coverage and Models. Data distribution is uneven across conditions: *keywuta* lacks NSMC examples, and the Vertex AI model was evaluated exclusively on dictionary definitions, preventing a full cross-register comparison. Moreover, the NSMC corpus represents only informal user reviews, which may not generalize to formal written or spoken registers. Additionally, frequency counts rely on automated morphological segmentation (Bareun, Mecab-ko) without exhaustive manual verification. Finally, Vertex AI’s elevated self-similarity values reflect the known anisotropy and reduced discriminative resolution of multilingual embeddings relative to monolingual models.

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Appendix

Execution Environment

All scripts were run in Python 3.14 with pandas 3.0, requests 2.34, transformers 5.12, and torch 2.12 (for the klue/bert-base and kcbert-base embedding extraction); notebooks were executed via `nbclient` and exported with `nbconvert`. Vertex AI embeddings (text-multilingual-embedding-002) were queried through Google Cloud’s Python SDK; NSMC frequencies were obtained via `mecab-ko` (Eunjeon Project, 2013) run in Google Cloud Shell. All corpora and tools used in this study (KoFREN, NSMC, STDICT, Woorimalsam, `mecab-ko`) are publicly released research resources, used here strictly for non-commercial academic purposes in accordance with their respective terms of use.

File Descriptions

- `dictionary_polysemy.ipynb/.html` — queries STDICT and Woorimalsam for each word, disambiguates homographs, and computes sense counts.
- `self_similarity.ipynb/.html` — computes contextual self-similarity (klue/bert-base, kcbert-base) across dictionary-example and NSMC usage sentences.
- `kofren_frequency.ipynb/.html` — computes log-frequency from KoFREN and cross-validates against NSMC.
- `data/dictionary_polysemy_results.csv` — STDICT/Woorimalsam sense counts per word.
- `data/kofren_frequency_results.csv` — raw counts and log-frequency from KoFREN.
- `data/nsmc_frequency_results.csv` — raw counts and log-frequency from NSMC (`mecab-ko`).
- `data/self_similarity_dict_examples_results.csv` — self-similarity scores on dictionary example sentences (klue, kcbert).
- `data/self_similarity_nsmc_results.csv` — self-similarity scores on NSMC sentences (klue, kcbert).
- `data/self_similarity_vertex_results.csv` — self-similarity scores from Vertex AI (text-multilingual-embedding-002).

Supplementary Materials

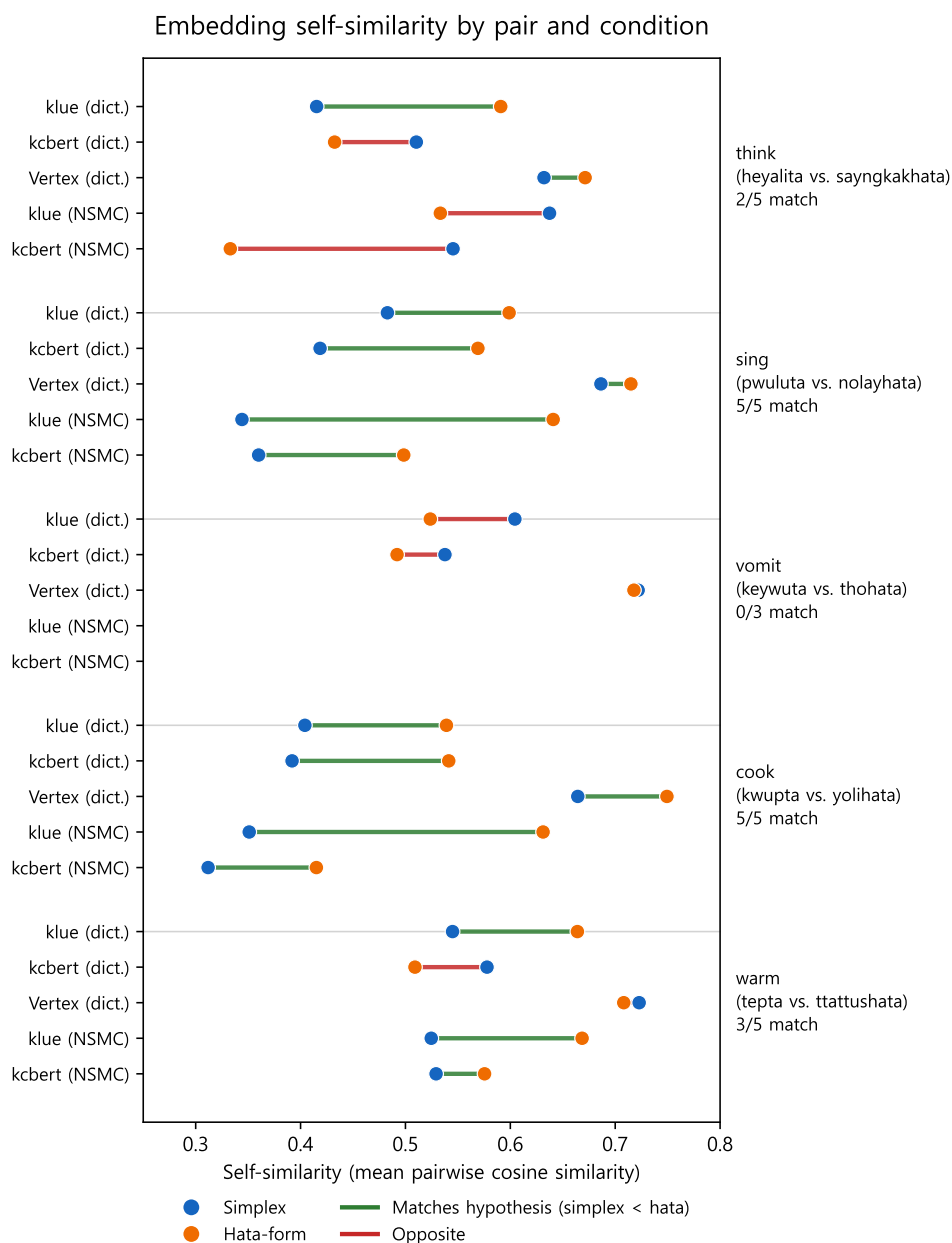


Figure 5: Self-similarity by condition and pair (dumbbell plot). Each row connects the simplex and *hata*-form values for one measurement condition; the connecting line is colored green where the simplex verb is less self-similar than its *hata*-counterpart (matching the hypothesis) and red where the reverse holds (see legend). Pair boundaries are marked by light gray horizontal lines.

Concept	Word	KoFREN ($\log_{10}$)	NSMC ($\log_{10}$)
think	<i>heyalita</i>	2.215	0.477
	<i>sayngkakhata</i>	4.887	3.166
sing	<i>pwuluta</i>	4.426	2.524
	<i>nolayhata</i>	3.251	1.505
vomit	<i>keywuta</i>	0.000	0.000
	<i>thohata</i>	3.286	1.342
cook	<i>kwupta</i>	3.912	1.000
	<i>yolihata</i>	3.471	0.699
warm	<i>tepta</i>	4.027	1.415
	<i>ttattushata</i>	4.119	2.336

Table 6: Log-transformed frequency by concept, word, and corpus. *keywuta* is unattested in KoFREN (raw count 0) and occurs only once in the entire 200,000-line NSMC corpus.